Cover letter for JMLR MLOSS submission

This submission is intended for the JMLR Machine Learning Open Source Software (MLOSS) track.

Manuscript title: *Sparse-conv Offline Perception with Hydra-MDP Inference: An Open Camera–LiDAR Planning Stack for NAVSIM*

1. Overlap with prior publication. This manuscript describes open-source software for training, evaluating, and deploying a camera–LiDAR fused BEV planner on NAVSIM, including an open sparse-convolution LiDAR runtime that replaces NVIDIA CUDA-BEVFusion's closed `libspconv` path, multi-head ONNX/TensorRT export (planning + detection + BEV segmentation), a one-frame example dataset for clone-and-run C++ inference, and a documented deploy workflow. No portion has appeared in a journal. Related public artifacts (CUDA-BEVFusion, Hydra-MDP/GTRS, NAVSIM, `spconv`) are cited and distinguished from our integration and deployment contributions.

2. Co-author consent. This is a single-author submission. The author confirms the manuscript and consents to its review by JMLR.

3. Conflicts of interest. None declared.

4. Open-source license. Apache License 2.0 (see `LICENSE` and `NOTICE`).

5. Project URL and version.

- Repository: `https://github.com/atharvsharma1998/hydra-mdp`
- Branch / version under review: `main` (software version `v0.1.0-mloss`)
- Deploy guide: `QUICKSTART.md`
- Checkpoints / ONNX (Google Drive): `https://drive.google.com/drive/folders/1hcnOaJvWhL3hzBxSSs`

6. Evidence of user community. The project is public with Issues enabled, Apache-2.0 licensing, install/deploy documentation (`QUICKSTART.md`, `docs/MODELS.md`), a shipped one-frame `deploy/example-data/` sample, and validated navtest PDM results. As a young single-maintainer release, external stars and forks are still accumulating; the submission emphasizes reproducibility (source archive, clone-and-run C++ path, Python–C++ parity).

7. Suggested action editors (no COI):

- Editors covering open-source ML software, robotics / autonomous driving systems, or efficient deep learning deployment.

8. Suggested reviewers (no COI) — by expertise:

- BEVFusion / multi-sensor BEV perception.
- `spconv` / sparse 3D convolution runtimes.
- NAVSIM / learned planning (Hydra-MDP, GTRS, PDM scoring).
- TensorRT / ONNX deployment systems.

9. Keywords: open source software; autonomous driving; bird's-eye view; sparse convolution; TensorRT; NAVSIM

10. Source code. Source is included in the submission archive and is publicly available at the project URL under Apache-2.0. Large checkpoints and optional ONNX bundles are distributed via Google Drive links documented in `docs/MODELS.md`; the archive includes a one-frame `example-data/` sample for C++ inference.

Corresponding author: Atharv Sharma (`atharv228.as@gmail.com`)

11. Funding disclosure. None.

12. Competing interests. None declared.

13. Proprietary dependencies. The documented C++ path requires NVIDIA CUDA and TensorRT (widely used in the ML systems community). The LiDAR SCN path does **not** require NVIDIA's closed `libspconv`; it uses open `spconv` 2.x. Full NAVSIM data is optional (train/eval only).